

Shadow Hand actuation: do not equate joints with controls

21 position servos -> 25 hand DOFs

17 joint actuators + four fixed-tendon actuators

MRL / FF distal actuation

J0 controls J2 + J1, with unit coefficients; the split is not fixed.

Scalar servo law

Force = clipped $k_p \times (\text{ctrl} - \text{actuator length})$. No actuator velocity damping.

Persistent preload

Blocked virtual target -> nonzero error -> actuator force -> contact wrench.